

CMP6200

Individual Undergraduate Project (FYP)

Project Interim Report

Real-Time Context-Aware Sign Language to Multilingual Speech Conversion System

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1. Introduction

The ability to communicate without barriers is fundamental to nearly every aspect of modern life, from health to education to employment. However, when it comes to the estimated 430 million people in the world who suffer from hearing impairment, communication remains a major challenge (WHO, 2023). Individuals who are deaf or hard of hearing communicate through sign language that is an entirely different language rather than being reduced forms of speaking languages (NIDCD, 2022). For example, Stokoe's pioneering structural analysis in 1960 resulted in the identification of American Sign Language (ASL) as an autonomous language.

Though efforts are still being made to create a more inclusive world, the communication gap between the signing and non-signing communities is still wide. Human interpreters are not always available, and earlier technological solutions relied on expensive hardware such as sensor-based gloves (Mitra & Acharya, 2007). However, with the advances in cheap RGB cameras and hand-pose estimation frameworks like MediaPipe (Zhang et al., 2020), software-based vision-based sign language recognition is now possible without the need for specialised equipment.

This project takes advantage of these advances by combining computer vision, natural language processing (NLP), and text-to-speech (TTS) synthesis into a single system. Recognition of the gestures in ASL by the system is performed via the use of the ordinary webcam that then reconstructs them in order to form sentences grammatically correct and translate them to spoken English and Nepali. Importance of integrating the speech output in Nepali language cannot be understated when taking into account that there are more than 40,000 deaf people living in Nepal who lack interpreters (World Federation of the Deaf, 2021).

1.1 Problem Statement

These concerns motivate our current research for the following reasons. Firstly, there is no affordable, real-time automated interpretation tool that can eliminate language barriers for deaf people in everyday interactions with hearing people. Second, there is a global dearth of qualified sign language interpreters, meaning that many communities, including Nepal's, are left without any professional recourse. Thirdly, although some automated tools have been designed, these glosses are grammatically incorrect and only understandable by professionals, making them practically useless. This project deals with all three issues in one integrated real-time multilingual pipeline.

2. Review of Existing Knowledge

2.1 Sign Language Recognition

The development of automatic sign language recognition (SLR) has been closely related to advances in machine learning. Mitra and Acharya (2007) made a distinction between skeletal hand models and appearance-based approaches in their early works. In subsequent studies on deep learning, CNN has further enhanced the static hand shape while LSTM model architecture has captured the dynamics of sign language in temporal modeling (Pigou et al., 2018; Hochreiter & Schmidhuber, 1997).

One major limitation in SLR research was the lack of large scale datasets to train deep models. This improved with datasets such as the ASL benchmark by Li et al. (2020) which contains over 21,000 labelled clips across 2,000 signs and the MS-ASL dataset by Joze and Koller (2019). These resources have made it possible for research to move from small laboratory vocabularies to large scale recognition systems.

Preprocessing also saw considerable progress with MediaPipe (Zhang et al., 2020) , which can extract 21 three-dimensional hand keypoints in real time with standard consumer hardware. The skeletal representation is resilient to changes in illumination, background, and even skin color, as opposed to the use of raw image data (Koller, 2020).

Transformer architectures originally proposed for NLP have recently been applied to sign recognition . Self-attention mechanisms that can model long-range sequence relationships without using recurrence were proposed by Vaswani et al. (2017). Following this, Camgöz et al. (2020) have demonstrated that Transformer encoder-decoder models outperform recurrent approaches on continuous sign language benchmarks, highlighting the increasing convergence between NLP and computer vision approaches in SLR research.

Approach / Tool	Application	Strengths / Limitations
CNN + LSTM	ASL gesture recognition	High accuracy on static signs; less effective on dynamic sequences
MediaPipe + Random Forest	Hand landmark-based SLR	Lightweight, real-time; limited vocabulary
Transformer (BERT/GPT)	Gloss-to-English NLP correction	Strong contextual grammar correction; needs domain-specific fine-tuning
3D-CNN	Continuous sign recognition	Captures temporal dynamics; computationally expensive
gTTS / Google Cloud TTS	Multilingual speech synthesis	High quality English TTS; Nepali support limited in open tools
WLASL Dataset	Word-level ASL dataset	10,000+ videos, 2000 signs; standard benchmark in recent literature

Figure 2.1: Table of Summary of Key Approaches in Sign Language Recognition Literature

2.2 Natural Language Processing for Grammar Correction

There is a difference in grammar between ASL and English, making it difficult to transform recognised signs into spoken language. In this case, ASL uses the topic-comment grammar style while English uses the subject-verb-object grammar form (NIDCD, 2022). ARE YOU GOING TO THE MOVIES TOMORROW? The word-by-word transformation will not be grammatically correct and therefore unusable in any other setting aside from sign language.

Stoll et al. (2020) approached this problem as a neural machine translation task, building sequence-to-sequence models trained on sign glosses and their corresponding English translations. Their attention-based model can be effective when it comes to putting words in order. Sequence correction has been improved further with large pre-trained models such as T5 (Raffel et al., 2020) and GPT-2 (Radford et al., 2019) fine-tuned on in-domain datasets. The quality of translations is often measured by the BLEU metric (Papineni et al., 2002) comparing the similarity of generated text with human reference translations.

2.3 Text-to-Speech Synthesis

The automatic speech synthesis has changed a lot in the last decade. Earlier statistical vocoders based on hidden Markov models were replaced by end-to-end neural architectures. WaveNet (van den Oord et al., 2016) showed that convolutional models working directly on audio waveforms could generate far more natural speech than previous systems. However, later models like Tacotron 2 (Shen et al., 2018) and FastSpeech (Ren et al., 2019) made speech synthesis more practical through the decoupling of prosody prediction from waveform generation.

Commercial APIs have neural voices available for multiple languages with response times suitable for real-time applications.

High-quality synthesis for low-resource languages remains challenging. For instance, Sitaula and Ghimire (2021) note that large curated speech datasets required for training advanced neural vocoders are not available for Nepali. Some commercial services have alleviated this limitation to some extent: Nepali voice models, available through commercial platforms such as Google Cloud TTS, provide adequate quality for assistive applications, but are still less expressive than models for high-resource languages.

2.4 End-to-End Integrated Systems

There have been a number of prototype systems that have tried to link sign recognition directly to speech output. Zafrulla et al. (2011) demonstrated an early example of depth-sensing cameras being used to recognise a limited vocabulary of ASL and generate speech. They note latency and vocabulary size restriction as significant barriers to practical use. Koller Survey (2020) demonstrated the same problems to persist: the great majority of systems are aimed at controlled environments, focused on isolated signs not continuous signing and lack grammatical error correction capabilities. It is important to mention that no open source system published was identified that combined continuous sign recognition, sentence level NLP correction, and multilingual speech output including Nepali.al speech output, including Nepali.

2.5 Critical Evaluation and Identification of Gaps

When evaluating the literature that has been reviewed above, there are three main gaps that can be identified. The first is coverage, since the vast majority of SLR systems are developed for limited vocabularies of isolated signs, and few are capable of processing the full continuum of connected, conversational signing in real time without the use of non-standard hardware (Koller, 2020). The second issue concerns the nature of the output generated; high performing recognizers generate their output in the form of raw gloss transcriptions, which can only be understood if the user knows the sign language (Zafrulla et al., 2011). Third is language coverage. There is a large population of deaf people in Nepal and a lack of interpreters, yet none of the sign-to-speech systems in the literature that we reviewed support Nepali audio output. Each gap alone would be sufficient to justify a research contribution. Collectively they define the scope of this project.

System / Study	Core Functionality	Hardware Requirement	Grammar Correction	Multilingual Speech Output	Real-Time Capability	Key Limitations
SignAll	AI-based sign language recognition and translation	Multiple cameras and specialised setup	Limited	Limited	Partial	Expensive deployment; limited accessibility for low-resource regions
Hand Talk	Converts text and speech into sign language avatars	Standard mobile device	N/A	Limited language support	Yes	Primarily supports speech/text-to-sign rather than sign-to-speech
Google Live Transcribe	Speech-to-text accessibility support	Standard mobile device	N/A	Multiple languages supported	Yes	Does not perform sign language recognition
Zafrulla et al. (2011) Kinect-based ASL System	ASL gesture recognition with speech synthesis	Kinect depth sensor	No	English only	Limited	Small vocabulary and dependence on specialised hardware
Camgöz et al. (2020) Transformer-Based SLR	Continuous sign language recognition and translation	High-performance GPU systems	Partial	English only	Research-focused	Computationally expensive; not designed for lightweight deployment
CNN/LSTM-based ASL Recognition Systems	Static and dynamic ASL sign classification	Webcam or video input	No	No	Partial	Outputs isolated glosses without contextual restructuring
Proposed System (This Project)	Real-time ASL recognition, NLP correction, and multilingual speech synthesis	Standard webcam and consumer hardware	Yes (Transformer-based)	English and Nepali speech output	Yes	Limited to 50+ sign vocabulary during initial implementation

Figure 2.2: Comparison Table

Table 2.2 demonstrates that existing sign language recognition systems primarily focus on isolated gesture classification or single-language output, with limited attention given to contextual grammar correction and multilingual accessibility. Furthermore, many existing approaches rely on specialised hardware or remain confined to research environments without real-time deployment considerations. The suggested system solves this problem by integrating ASL recognition from webcams, sentence reconstruction based on Transformer, and multilingual speech synthesis in one pipeline.

2.6 Project Justification

By building a pipeline that chains a CNN-LSTM recogniser, a domain-adapted T5 grammar correction module, and a multilingual TTS component, this project directly addresses each identified gap. Operating on nothing more than a webcam and a mid-range laptop, it avoids the hardware barriers that have limited earlier work. The inclusion of Nepali output gives the tool immediate relevance for an underserved community that currently has no equivalent technological resource. The performance goals of at least 85% recognition rate and overall latency of less than 1.5 seconds are specified with respect to established benchmarks from previous literature and should represent the limit of practical usability rather than purely academic satisfaction.

3. Scope, Objective, and Goals of the Project

3.1 Project Aim

This project's goal is to create and validate a real-time AI-based system that can convert American Sign Language gestures captured through a consumer webcam into grammatically correct English and Nepali spoken sentences. It is accomplished through the following steps one after another: Gesture recognition with the help of deep learning, sentence reconstruction based on NLP and contextual awareness, and finally multilingual speech synthesis. The larger motivation is to expand access to communication for deaf and hard of hearing people in situations where there are no qualified interpreters available.

3.2 Project Objectives

The five objectives that have been defined below are SMART objectives and are presented in Table 3.1:

Obj.	Description	Target	Success Metric
O1	Develop CNN+LSTM ASL recognition model for ≥ 50 signs at $\geq 85\%$ accuracy	Week 20	Accuracy, Precision, Recall, F1-score
O2	Implement T5-based NLP correction module improving fluency by $\geq 20\%$ over baseline	Week 24	BLEU score vs baseline
O3	Integrate English + Nepali TTS with end-to-end latency < 1.5 seconds	Week 26	Latency (ms) across 50 sequences
O4	Unify all modules into a real-time webcam-accessible pipeline	Week 28	End-to-end accuracy & latency on 100 sequences
O5	Evaluate system with quantitative metrics and usability testing ($\geq 80\%$ satisfaction)	Week 32	SUS score, confusion matrix, BLEU, latency

Figure 3.1: Table of SMART Project Objectives Summary

- Objective 1:** To develop a hybrid CNN-LSTM classifier which will be capable of recognising at least 50 ASL signs with an accuracy of not less than 85%. Success will be measured via precision, recall, F1-score, and confusion matrix analysis. Completion target: Week 20.
- Objective 2:** Develop and optimize a T5-based seq2seq model that reconfigures sign glosses into grammatically correct English sentences. It must improve the BLEU metric score by at least 20 percent relative to the unmodified word-sequence baseline. Target completion date: Completion target: Week 24.

3. **Objective 3:** Implement the English and Nepali speech synthesis subsystems in the pipeline such that the time delay between the end of gesture recognition and the beginning of audio playback is no more than 1.5 seconds; validated via a battery of 50 test gestures.
4. **Objective 4:** Bring all three subsystems together into a single continuous pipeline that does not need any additional hardware beyond a regular webcam. Test the resulting system on a set of 100 held-out American Sign Language test gestures.
5. **Objective 5:** Carry out a formal evaluation of the pipeline, employing various measures like accuracy, BLEU score, latency, and confusion matrix in conjunction with a user experience study using a minimum of ten users comprising deaf, interpreters, and hearing participants. The target SUS score is ≥ 80 . Completion Target: Week 32.

3.3 Project Scope

Scope: Recognition of ASL gestures based on a vocabulary consisting of at least 50 signs; input using only a webcam; grammar correction for English in context-sensitive fashion; speech output in both English and Nepali; real-time operation on off-the-shelf computers; quantitative benchmarking and participant usability testing.

Not included: Any sign language besides ASL (for instance, BSL or Nepali Sign Language); speech output in languages besides English and Nepali; use of mobile platforms or embedded processors; recognition of non-manual information like facial expressions or mouth actions; and automatic adjustment to user-specific signing styles without re-training the base model.

4. Project Design

4.1 System Architecture

The proposed system is organized as a linear pipeline with four separate layers, each of which can be tested in isolation. The modular nature is that the various parts may be separately designed, tested, and connected to one another in order to reduce the chances of total failure upon integration.

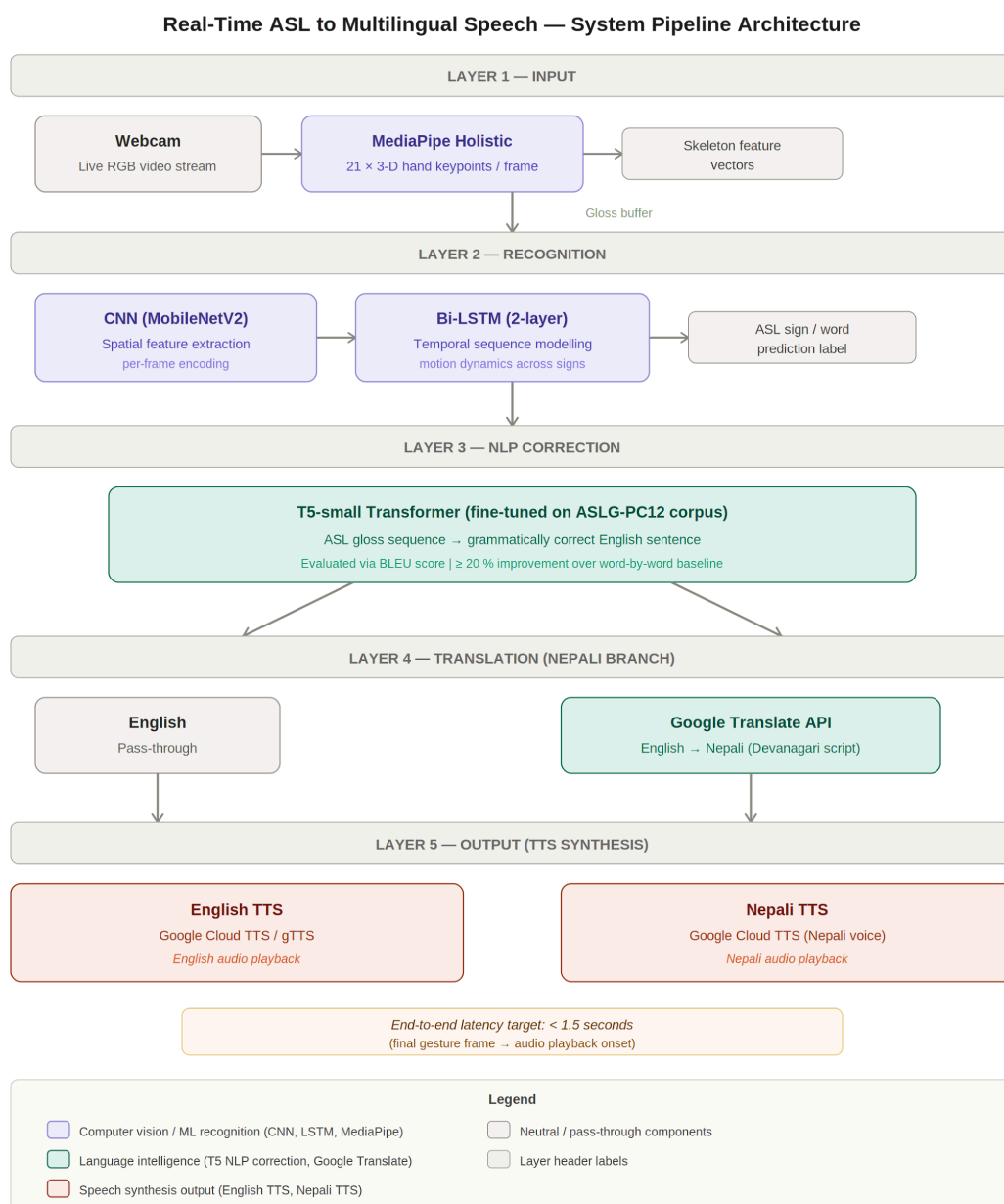


Figure 4.1: High-Level System Pipeline Architecture

4.2 Design and Methods

Development is organised into five sequential phases: data preparation, model training, module integration, evaluation, and final reporting.

Phase 1 - Data Preparation. Two openly licensed ASL datasets supply the training material. The first one would be an image dataset of 87,000 samples labeled into 29 different configurations of the hands, useful for training static sign recognition systems (Grassi & Akash, 2018). The second is the WLASL video corpus (Li et al., 2020), which covers a vocabulary of dynamic signs. Before training, MediaPipe Holistic (Zhang et al., 2020) uses the input frames to produce 21 three-dimensional hand keypoints, transforming pixel images into a compact skeletal representation resilient to changes in lighting conditions and user variance.

Dataset Justification

The choice of training datasets was based on the evaluation of multiple public ASL corpora that satisfy certain criteria: scale, quality of annotations, licence conformity with academic research requirements, and suitability for two subtasks that should be solved in ASL recognition: static hand shape recognition and sign sequence recognition.

ASL Alphabet Dataset (Grassi & Akash, 2018). For static hand shape recognition, the most suitable dataset among freely accessible corpora is Kaggle ASL Alphabet Dataset. It offers 87,000 annotated images across 29 classes that can be used to train a CNN until it converges, as there is no need for excessive augmentation of such amount of data. Some other datasets (smaller academic lab datasets) were taken into consideration but not accepted since they offer less than 5,000 images in total, which makes it impossible to train a deep model without a risk of overfitting. An additional factor in favor of the dataset is its licence.

WLASL - Word-Level American Sign Language (Li et al., 2020). For the task of dynamic sign recognition, WLASL is at present the most extensive publicly accessible video database of word-level ASL, providing over 21,000 annotated videos with 2,000 unique signs. Its closest competitor is MS-ASL (Joze & Koller, 2019), which is comparable in size but has stricter distribution requirements making its use difficult for research purposes. Another advantage of WLASL is the train, validation, and test splits, making the evaluation more credible in terms of comparability to existing literature. Despite being not representative of all signing styles and demographic groups equally, which is listed as a drawback in the risk register, it is representative of the widest range possible using publicly available data.

ASLG-PC12 Corpus. In order to train the NLP grammar correction module, the ASLG-PC12 corpus contains matched gloss and sentence example pairs in English based on actual signing. There is no such other publicly licensed parallel corpus of gloss into English translation available in a larger scale, which makes this the only existing parallel resource that could be used for fine-tuning T5 on the gloss to English translation task.

The dataset for Nepali Sign Language is not available. It is important to note that the labelled dataset of Nepali Sign Language (NSL) is not available at the time of writing. Consequently, this project concentrates on the recognition of ASL and works only on Nepali language during the synthesis stage. The output in corrected English form is converted into Nepali audio using TTS. This limitation is not by chance but due to the scope of this project mentioned in Section 3.3.

Phase 2 - Train the recognition model. The recogniser uses a bidirectional two-layer LSTM architecture in combination with a MobileNetV2 CNN due to its effective accuracy-to-parameter tradeoff. The CNN produces spatial feature vectors, one per frame. The LSTM takes input in the form of vectors in time order and trains on how to recognize differences between various motions. The training involves using the Adam optimizer with 80/10/10 splits and categorical cross entropy. Evaluation metrics include accuracy, precision, recall, F1-score, and confusion matrix to reveal errors per class.

Phase 3 - Grammar Correction. A T5-small model (Raffel et al., 2020) was pre-trained on a general large text corpus and fine-tuned on a specially selected dataset of pairs of ASL glosses and corresponding English sentences from the ASLG-PC12 corpus, along with hand-curated samples. The performance of the grammar correction component is measured by BLEU, which evaluates the output produced with and without the component against a human-made translation.

Phase 4 - Incorporate multilingual TTS. English speech is generated using Google Cloud Text-to-Speech API. We have kept gTTS as a fallback, and for Nepali, we use the same platform with its Nepali voice model. Then the corrected English sentence is translated into Nepali using the Google Translate API and then synthesized. The performance indicator for this phase is the end-to-end latency, which is the time interval between the last recognised gesture frame and the start of audio output.

Phase 5 - Conducting User Studies & Evaluation. The user study shall involve at least ten subjects, three different types of users being the Deaf and Hard of Hearing Signers, professional interpreters, and hearing non-signers. Each subject then performs a series of assigned tasks followed by the System Usability Scale questionnaire. Information will be gained from qualitative data gathered from task observations and interviews that may not necessarily be revealed by the SUS scores alone.

4.3 Justification of Methods

Selection of a tool is always a compromise. Selection of a CNN-LSTM architecture over the Transformer is made for the recognizer because the CNN-LSTM architecture provides an acceptable level of accuracy on a 50-sign dictionary with fewer parameters to keep latency under the target value. Transformer-based recognition will be revisited as a potential upgrade once the pipeline is operational. In particular, we like to work with MediaPipe rather than pixels, as the output of skeletons gets rid of all the noise associated with background and lighting and significantly reduces the number of features (Koller, 2020). The encoder-decoder architecture of T5 is naturally suited to the gloss-to-sentence mapping task in the NLP correction stage. GPT-2 is a decoder that only generates text and is not designed to take a full input sequence and transform it (Raffel et al., 2020). We choose Cloud TTS over a custom vocoder, as training a neural TTS system for Nepali from scratch would require labelled speech resources and development time far beyond the scope of this project, but the commercial APIs offer acceptable audio quality for an assistive prototype. The SUS test is used to measure the usability of a product since it is a scale consisting of ten items which can be scored within five minutes. It is not language or domain-specific and has well-defined norms (Brooke, 1996).

4.4 Project Timeline

The table below maps each major task to its scheduled project phase. Phases overlap in part where dependency chains allow.

Task	Timeline	Ph1W k1-6	Ph2W k7-10	Ph3W k11-20	Ph4W k18-24	Ph5W k22-26	Ph6W k25-28	Ph7W k28-34
Literature Review & Proposal	Week 1–6	■						
Dataset Collection & Pre-processing	Week 5–10		■					
ASL Recognition Model (CNN+LSTM)	Week 9–20			■				
NLP Context-Correction Module (T5)	Week 18–24				■			
Multilingual TTS Integration	Week 22–26					■		
Pipeline Integration & Testing	Week 25–28						■	
Usability Testing & Evaluation	Week 28–32							■
Final Report & Submission	Week 30–34							■

Figure 4.2: Project Gantt Chart

5. Feasibility, Risk, and Ethical Considerations

5.1 Feasibility

The project is technically feasible within its 32-week time frame for four reasons. First, all the required datasets and frameworks such as MediaPipe, TensorFlow, and PyTorch are free and open-source. Second, Google Cloud TTS and Translate APIs have free tiers that are sufficient for development and evaluation. Thirdly, the training for the model will take place on Google Colab Pro with GPU computing power of NVIDIA A100 and V100 instances and without the need for any special hardware on the user's end. Lastly, the developer possesses the required skills in Python programming, machine learning algorithms, and web applications and has ongoing guidance from their supervisor.

In terms of architecture, it is much simpler now too. The three major modules, Recognition, Correction, and Synthesis, are developed separately, tested, and eventually incorporated, making it easier to debug the code when needed. Moreover, the scheduling of the project on the Gantt chart allows some slack between milestones.

5.2 Risk Analysis and Mitigation

There are six risks associated with a high level of probability/impact. Each risk along with its probability, impact, and proposed method of mitigation is listed on Table 5.1.

Potential Risk	Likelihood	Impact	Mitigation Strategy
Model accuracy below 85% on ASL recognition	Medium	High	Use well-established public datasets (WLASL, ASL Alphabet). Apply data augmentation and cross-validation. Fall back to a simpler CNN baseline.
NLP correction module underperforms on ASL gloss sequences	Medium	High	Fine-tune a pre-trained T5 model on gloss-to-English pairs. Use BLEU score benchmarking at each iteration to detect regression early.
Nepali TTS engine produces unnatural or unintelligible output	High	Medium	Evaluate multiple Nepali TTS engines early in the project. Maintain English-only output as a fallback while Nepali is refined.
System latency exceeds 1.5-second threshold in real-time pipeline	Medium	High	Profile each pipeline stage separately. Optimise bottlenecks via model quantisation or batching. Test on standard consumer hardware from Week 20.
Insufficient usability testing participants from deaf communities	Medium	High	Begin participant recruitment in Week 26. Partner with local disability organisations. Use online deaf community forums as backup recruitment channels.
Dataset licensing restrictions prevent use in research	Low	High	Verify licences before committing to datasets. Prioritise fully open datasets (WLASL, Kaggle ASL Alphabet). Document all licensing decisions.

Figure 5.1: Risk Assessment and Mitigation Strategies

5.3 Ethical Considerations

Because the project involves webcam footage of human participants and members of a vulnerable community, ethical responsibilities are carefully considered at all stages of the study. The following safeguards are applicable to all relevant aspects of the project.

Consent informed. Prior to any data collection being initiated, participants will be offered an information sheet that will include details about the aim of the research, type of data to be gathered, and also where it will be stored, with participant's option of withdrawing at any point.

Anonymisation of data. As soon as they are recorded, webcam footage and gesture recordings are anonymized. Performance data will not be saved together with the demographic data. The data will all be saved in password-protected institutional storage and will be securely deleted upon project completion.

Well-being of participants. Study sessions are 45 minutes duration, with no pressure of time. If required, arrangements will be made to facilitate interpreter services or textual communication.

Data ethics. Training datasets are used solely within the context of their academic research

licences. Third-party information is not personally identifiable information, and proprietary material is not used without permission.

Fairness (Algorithmic). Recognition accuracy will be analyzed in relation to factors like age, gender, and skin color. Computer vision systems can be biased if they are trained on non-representative data (Schroeder et al., 2020). Any disparities identified will be clearly reported in the final study.

Social Impact and Transparency. This is a research prototype, and the limitations and failure cases of the system will be transparently reported. Participants and the general public will be made aware that this tool is not meant to replace professional human translators and is not suitable for use in scenarios where there are high risks involved.

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